EE4-66 Topics in Large Dimensional Data Processing

Instructions for Candidates

Answer all three questions. Total marks: 75.

1. (Linear Inverse Problem)

   (a) Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a flat matrix, i.e., $m < n$.

      i Consider the full singular value decomposition (SVD) of $\boldsymbol{A}$ in the form of $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$. Identify the dimensions and properties of the matrices $\boldsymbol{U}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{V}$, respectively. [3]

      ii Let r be the (row) rank of $\boldsymbol{A}$. Suppose that $r \leq m < n$. The compact SVD of $\boldsymbol{A}$ is given by $\boldsymbol{A} = \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^T$.

         A. Identify the dimensions of the matrices $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$, respectively. [2]

         B. Describe the relationship between $\boldsymbol{U}_r, \boldsymbol{\Sigma}_r, \boldsymbol{V}_r$ and $\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$. [2]

      iii Decompose the matrices $\boldsymbol{A}\boldsymbol{A}^T$ and $\boldsymbol{A}^T\boldsymbol{A}$ and write your results in terms of $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$. [2]

      iv Define the pseudo-inverse of $\boldsymbol{A}$, denoted by $\boldsymbol{A}^\dagger$, using $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$. [2]

      v For a given arbitrary $\boldsymbol{y} \neq \boldsymbol{0}$ and $\boldsymbol{y} \in \text{range}(\boldsymbol{U}_r)$, it is clear that the set defined as

$$\mathcal{X} := \{\boldsymbol{x} : \ \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}\}$$

      is not empty.

         A. Let $\boldsymbol{x}_0 = \boldsymbol{A}^\dagger\boldsymbol{y}$ where $\boldsymbol{A}^\dagger$ is the psuedo-inverse of $\boldsymbol{A}$ defined in 1.(a)-iv. Show that $\boldsymbol{x}_0 \in \mathcal{X}$. [2]

         B. Define

$$\mathcal{X}_\perp := \{\boldsymbol{x}_\perp : \ \boldsymbol{A}\boldsymbol{x}_\perp = \boldsymbol{0}\}.$$

         It is clear that any $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{x}_\perp$, where $\boldsymbol{x}_\perp \in \mathcal{X}_\perp$, is a solution of $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$.
         Find the dimension of the subspace $\mathcal{X}_\perp$ and an orthonormal basis of it via the SVD of $\boldsymbol{A}$. [2]

         C. It is a well-known fact that for all $\boldsymbol{x} \in \mathcal{X}$, it holds $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{x}_\perp$ for some $\boldsymbol{x}_\perp \in \mathcal{X}_\perp$.
         Prove that

$$\|\boldsymbol{x}_0\|_2^2 = \min_{\boldsymbol{x} \in \mathcal{X}} \|\boldsymbol{x}\|_2^2.$$

         [2]

(b) Let

$$\boldsymbol{A} = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 1 & 1 & -1 & -1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

i Compute $\boldsymbol{A}\boldsymbol{A}^T$. [2]

ii Let $\boldsymbol{v} = \frac{1}{2}[1, -1, 1, -1]^T$. Compute $\|\boldsymbol{v}\|_2$ and $\boldsymbol{A}\boldsymbol{v}$. [2]

iii The *full* SVD of $\boldsymbol{A}$ can be written as $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$. Find $\boldsymbol{U}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{V}$ numerically. [2]

iv Find the pseudo-inverse of $\boldsymbol{A}$ numerically, denoted by $\boldsymbol{A}^\dagger$. [3]

v Let $\boldsymbol{y} = [4, 2, 1]^T$. Find $\boldsymbol{x} = \boldsymbol{A}^\dagger\boldsymbol{y}$ numerically. [2]

vi Let $\boldsymbol{x}'$ satisfy that $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}'$ and $\|\boldsymbol{x}'\|_0 = 1$. Find $\boldsymbol{x}'$ numerically. [2]

(Total marks: 30)

2. (Greedy Algorithms for Sparse Inverse Problems) Consider greedy algorithms to solve the following sparse inverse problem:

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 \ \text{s.t.} \ \|\boldsymbol{x}\|_0 \leq S,$$

where $\boldsymbol{y} \in \mathbb{R}^m$, $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{x} \in \mathbb{R}^n$, and $0 < S \ll n$ is a given positive integer. We assume that all the columns of $\boldsymbol{A}$ are normalised, i.e., $\|\boldsymbol{a}_j\|_2 = 1$ where $\boldsymbol{a}_j$ is the j-th column of $\boldsymbol{A}$.

(a) The Orthogonal Matching Pursuit (OMP) algorithm is stated in Algorithm 2.1.

Algorithm 2.1 OMP Algorithm

1: **Input**: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
2: **Initialisation**: $\boldsymbol{x} = \boldsymbol{0}$, $\mathcal{T}^0 = \phi$, and $\boldsymbol{y}_r = \boldsymbol{y}$.
3: **for** $l = 1, 2, \cdots, S$ **do**
4: $\quad i_l = ??$ ▷ Find a new index
5: $\quad \mathcal{T}^l = \mathcal{T}^{l-1} \bigcup \{i_l\}$ ▷ Expand the support set $\mathcal{T}$
6: $\quad \boldsymbol{x}_{\mathcal{T}^l} = ??$ ▷ Update $\boldsymbol{x}$
7: $\quad \boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

i Lines 4 and 6 in OMP are incomplete. Complete them. [2]

ii Prove that generally speaking $i_l \notin \mathcal{T}^{l-1}$. [4]

iii The performance guarantee of OMP states that if the mutual coherence of the matrix $\boldsymbol{A}$ satisfies

$$\mu(\boldsymbol{A}) \leq \frac{1}{2S},$$

then OMP guarantees to recover an arbitrary S-sparse signal $\boldsymbol{x}_0$ exactly.

A. State the definition of mutual coherence $\mu(\boldsymbol{A})$. [1]

B. Let $\boldsymbol{x}_0$ be the ground-truth signal that generates $\boldsymbol{y}$, i.e., $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0$. Let $\mathcal{T}_0 = \text{supp}(\boldsymbol{x}_0)$ and $|\mathcal{T}_0| \leq S$. Show that for all $i \notin \mathcal{T}_0$, $|\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| \leq \mu\sqrt{S}\,\|\boldsymbol{x}_0\|_2$. [3]
Hint: Cauchy-Schwartz inequality $|\langle \boldsymbol{u}, \boldsymbol{v}\rangle| \leq \|\boldsymbol{u}\|_2 \|\boldsymbol{v}\|_2$ is needed.

(b) The Subspace Pursuit (SP) algorithm is stated in Algorithm 2.2.

i Complete Lines 4, 5, 6, and 7 of the SP algorithm. [2]

ii In the SP algorithm, H_S denotes the hard thresholding function with sparsity S. Find $H_3([-9, 8, -7, 6, -5]^T)$. [2]

Algorithm 2.2 SP Algorithm

Input: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
Initialisation: $\mathcal{T}^0 = \text{supp}(H_S(\boldsymbol{A}^T\boldsymbol{y}))$, and $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^0}\boldsymbol{A}^{\dagger}_{\mathcal{T}^0}\boldsymbol{y}$.
while $l = 1, 2, \cdots$ until stop criteria hold **do**
    $\tilde{\mathcal{T}}^l = \mathcal{T}^{l-1} \bigcup ??$ ▷ Expand support
    Set $\boldsymbol{b}$ by ?? ▷ Estimate a $2S$-sparse signal
    Set $\mathcal{T}^l =??$ ▷ Shrink support
    Set $\boldsymbol{x}$ by ?? ▷ Estimate an S-sparse signal
    $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

iii The performance guarantees of SP is obtained through the so called restricted isometry property (RIP) and restricted isometry constant (RIC). State the definitions of RIP and RIC. [4]

iv Consider two submatrices of $\boldsymbol{A}$, given by $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$ where $\mathcal{I}, \mathcal{J} \subset \{1, 2, \cdots, n\}$, $\mathcal{I} \bigcap \mathcal{J} = \phi$, and $|\mathcal{I}| = |\mathcal{J}| = S$. RIP suggests near-orthogonality of $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$. Find the mathematical expression to state the near-orthogonality of $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$ using singular values and RIC. Proof is not needed here. [2]

(Total marks: 20)

3. (Convex Optimisation, and Sparse Inverse Problems)

(a) Consider the following constrained optimisation problem

$$\min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{x}\|_2^2 \text{ s.t. } \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}, \tag{3.1}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ is the unknown variable vector to be optimised, $\boldsymbol{y} \in \mathbb{R}^m$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ are given.

i Let $\boldsymbol{v}$ be the corresponding Lagrange multiplier vector. Specify the dimension of $\boldsymbol{v}$ and find the Lagrangian of (3.1), denoted by $L(\boldsymbol{x}, \boldsymbol{v})$. [2]

ii Find the corresponding dual function $g(\boldsymbol{v})$. Solve the optimisation problem involved and obtain the closed form of $g(\boldsymbol{v})$. [2]

iii Find the form of the corresponding dual problem. Solve it by finding the optimal $\boldsymbol{v}^\star$. [2]

iv Find the closed form of the optimal solution $\boldsymbol{x}^\star$ of (3.1). [2]

(b) Consider the Lasso problem

$$\min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda \|\boldsymbol{x}\|_1, \tag{3.2}$$

where $\lambda > 0$ is a regularization constant.

i Consider an easier problem

$$\min_x f(x) := \min_x \frac{1}{2}(y - x)^2 + \lambda |x|,$$

where $x, y \in \mathbb{R}$. Find the derivative (or sub-differential) $\partial_x f(x)$. Then find the optimal solution $x^\star = \eta_\lambda(y)$ of the above scalar Lasso problem. [4]

In the literature, $\eta_\lambda(y)$ is referred to as the soft thresholding function.

ii Define

$$g(\boldsymbol{x}) := \frac{1}{2} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

For a fixed point $\boldsymbol{x}^k$, $g(\boldsymbol{x})$ can be approximated by

$$g(\boldsymbol{x}) \approx \frac{1}{2} \left\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^k\right\|_2^2 + \left\langle \nabla g\left(\boldsymbol{x}\right)|_{\boldsymbol{x}^k}, \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2t_k} \left\|\boldsymbol{x} - \boldsymbol{x}^k\right\|_2^2, \tag{3.3}$$

for some properly chosen $t_k > 0$ when $\boldsymbol{x}$ is sufficiently close to $\boldsymbol{x}^k$.

Use the approximation (3.3) to derive the Iterative Shrinkage Thresholding (IST) algorithm to solve the Lasso problem (3.2). Use the soft thresholding function in your final solution. [3]

(c) Consider the constrained optimisation problem

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{x}\|_1 \ \text{s.t.} \ \boldsymbol{y} = \boldsymbol{D}\boldsymbol{x}, \tag{3.4}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{y} \in \mathbb{R}^m$, and $\boldsymbol{D} \in \mathbb{R}^{m \times n}$.

We will derive an ADMM algorithm to solve (3.4).

Generally speaking, ADMM is designed to solve convex optimisation problem in the following standard form

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{z}} \ f(\boldsymbol{x}) + g(\boldsymbol{z}) \\ &\text{s.t.} \ \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} = \boldsymbol{c}. \end{aligned} \tag{3.5}$$

In high level, ADMM solves the above optimisation problem by solving much easier sub-problems iteratively.

i Consider the optimisation problem

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{z}} \ \|\boldsymbol{z}\|_1 \\ &\text{s.t.} \ \boldsymbol{D}\boldsymbol{x} = \boldsymbol{y}, \ \text{and} \ \boldsymbol{x} = \boldsymbol{z}. \end{aligned} \tag{3.6}$$

Clearly (3.6) and (3.4) are equivalent. The advantage of (3.6) is that sub-problems in ADMM will be easy to solve. Specify f, g, $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ when mapping (3.6) directly to the ADMM standard form (3.5). [3]

ii Find the augmented Lagrangian $L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v})$. Use $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ to simplify the notations. Find the dimension of the Lagrange multiplier vector $\boldsymbol{v}$. [3]

iii There are three steps in each iteration of ADMM:

$$\boldsymbol{x}^{k+1} = \arg\min_{\boldsymbol{x}} \ L_\rho(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{v}^k), \tag{3.7}$$

$$\boldsymbol{z}^{k+1} = \arg\min_{\boldsymbol{z}} \ L_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{v}^k), \tag{3.8}$$

$$\boldsymbol{v}^{k+1} = \boldsymbol{v}^k + \rho(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z}^{k+1} - \boldsymbol{c}).$$

Find the closed-form solutions of (3.7) and (3.8) respectively. [4]

(Total marks: 25)

Solution to Problem 1.

(a) i $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ where $\boldsymbol{U} \in \mathbb{R}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ are unitary, i.e., $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$ and $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$, and $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ is diagonal. (The convention is that the diagonal elements of $\boldsymbol{\Sigma}$ are sorted from the largest to the smallest.) [3]

ii A. $\boldsymbol{U}_r \in \mathbb{R}^{m\times r}$, $\boldsymbol{\Sigma}_r \in \mathbb{R}^{r\times r}$, and $\boldsymbol{V}_r \in \mathbb{R}^{r\times n}$. [2]

B. $\boldsymbol{U}_r = \boldsymbol{U}_{:,1:r}$, $\boldsymbol{\Sigma}_r = \boldsymbol{\Sigma}_{1:r,1:r}$, and $\boldsymbol{V}_r = \boldsymbol{V}_{:,1:r}$. [2]

iii $\boldsymbol{A}\boldsymbol{A}^T = \boldsymbol{U}_r\boldsymbol{\Sigma}_r^2\boldsymbol{U}_r^T$, and $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}_r\boldsymbol{\Sigma}_r^2\boldsymbol{V}_r^T$. [2]

iv $\boldsymbol{A}^\dagger = \boldsymbol{V}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{U}_r^T$. (As $\boldsymbol{\Sigma}$ is a diagonal matrix with nonzero diagonal elements, $\boldsymbol{\Sigma}_r^{-1}$ is well defined and easy to compute.) [2]

v A. It suffices to show that $\boldsymbol{A}\boldsymbol{x}_0 = \boldsymbol{y}$. $\boldsymbol{y} \in \text{range}(\boldsymbol{U}_r)$ implies that $\boldsymbol{y} = \boldsymbol{U}_r\boldsymbol{c}$ for some nonzero vector $\boldsymbol{c} \in \mathbb{R}^r$.
Now we compute $\boldsymbol{A}\boldsymbol{x}_0$:

$$\begin{aligned}\boldsymbol{A}\boldsymbol{x}_0 &= \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{y}\\ &= (\boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^T)(\boldsymbol{V}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{U}_r^T)(\boldsymbol{U}_r\boldsymbol{c})\\ &= \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{I}_r\boldsymbol{\Sigma}_r^{-1}\boldsymbol{I}_r\boldsymbol{c}\\ &= \boldsymbol{U}_r\boldsymbol{c} = \boldsymbol{y}.\end{aligned}$$

Hence $\boldsymbol{x}_0 \in \mathcal{X}$. [2]

Comments

A common mistake here was "$\boldsymbol{U}_r\boldsymbol{U}_r^T = \boldsymbol{I}_m$", which is wrong. For example, let

$$\boldsymbol{U}_r = \begin{bmatrix}1\\0\\0\end{bmatrix}.$$

On the other hand, $\boldsymbol{U}_r^T\boldsymbol{U}_r = \boldsymbol{I}_r$ always hold.
The matrices $\boldsymbol{U}_r\boldsymbol{U}_r^T$ and $\boldsymbol{U}_r^T\boldsymbol{U}_r$ have the same rank but very different sizes!

B. The subspace $\mathcal{X}_\perp$ has dimension $n - r$. One orthonormal basis of $\mathcal{X}_\perp$ is given by $\boldsymbol{V}_{:,r+1:n}$ where $\boldsymbol{V}$ is the right singular vector matrix of $\boldsymbol{A}$. [2]

C. For all $\boldsymbol{x} \in \mathcal{X}$, it holds that

$$\begin{aligned}\|\boldsymbol{x}\|_2^2 &= \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}_0 + \boldsymbol{x}_\perp, \boldsymbol{x}_0 + \boldsymbol{x}_\perp \rangle \\ &= \|\boldsymbol{x}_0\|_2^2 + 2\langle \boldsymbol{x}_0, \boldsymbol{x}_\perp \rangle + \|\boldsymbol{x}_\perp\|_2^2 \\ &\overset{(a)}{=} \|\boldsymbol{x}_0\|_2^2 + \|\boldsymbol{x}_\perp\|_2^2 \\ &\geq \|\boldsymbol{x}_0\|_2^2,\end{aligned}$$

where (a) follows from the orthogonality between $\boldsymbol{x}_0$ and $\boldsymbol{x}_\perp$, and the last inequality becomes equality if and only if $\boldsymbol{x}_\perp = \boldsymbol{0}$. Hence $\|\boldsymbol{x}_0\|_2^2 = \min_{\boldsymbol{x} \in \mathcal{X}} \|\boldsymbol{x}\|_2^2$. [2]

(b) i It is straightforward to verify that

$$\boldsymbol{A}\boldsymbol{A}^T = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

[2]

ii It is elementary to see that $\|\boldsymbol{v}\|_2 = 1$ and $\boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}$. [2]

iii It can be verified that

$$\begin{aligned}\boldsymbol{A} &= \begin{bmatrix} 2 & 2 & 2 & 2 \\ 1 & 1 & -1 & -1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\boldsymbol{U}} \underbrace{\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\boldsymbol{\Sigma}} \underbrace{\frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}}_{\boldsymbol{V}^T}\end{aligned}$$

[2]

Comments

Some students tried the way of computing singular values of $\boldsymbol{A}$ via computing the eigenvalues of $\boldsymbol{A}\boldsymbol{A}^T$ and got the correct answer. At the same time, a much easier way is a careful inspection. The matrix $\boldsymbol{A}$ in this question has been designed in such a way that tedious computations can be avoided.

iv

$$\begin{aligned}\boldsymbol{A}^{\dagger} &= \boldsymbol{V}_r \boldsymbol{\Sigma}_r^{-1} \boldsymbol{U}_r \\ &= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.\end{aligned}$$

[3]

v

$$\begin{aligned}\boldsymbol{x} = \boldsymbol{A}^{\dagger}\boldsymbol{y} &= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 3 \\ 1 \\ -1 \\ 1 \end{bmatrix}.\end{aligned}$$

[2]

vi By inspection, it is clear that $\boldsymbol{x}' = [2, 0, 0, 0]^T$. [2]

Comments

Similar to solving many math puzzles, the best way to get the correct answer is "by inspection". A fraction of students was able to solve this question by using this method.

Solution to Problem 2.

(a) i The complete OMP algorithm is stated in Algorithm 2.3.

Algorithm 2.3 OMP Algorithm

1: **Input**: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
2: **Initialisation**: $\boldsymbol{x} = \boldsymbol{0}$, $\mathcal{T}^0 = \phi$, and $\boldsymbol{y}_r = \boldsymbol{y}$.
3: **for** $l = 1, 2, \cdots, S$ **do**
4: $\quad i_l = \arg\max_i |\langle \boldsymbol{a}_i, \boldsymbol{y}_r \rangle|$ ▷ Find a new index
5: $\quad \mathcal{T}^l = \mathcal{T}^{l-1} \bigcup \{i_l\}$ ▷ Expand the support set $\mathcal{T}$
6: $\quad \boldsymbol{x}_{\mathcal{T}^l} = \boldsymbol{A}_{\mathcal{T}^l}^{\dagger} \boldsymbol{y}$ ▷ Update $\boldsymbol{x}$
7: $\quad \boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

[2]

ii It is clear that

$$\begin{aligned} \boldsymbol{y}_r &= \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^{l-1} \\ &= \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{x}_{\mathcal{T}^{l-1}}^{l-1} \\ &= \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{A}_{\mathcal{T}^{l-1}}^{\dagger} \boldsymbol{y}. \end{aligned}$$

Hence,

$$\begin{aligned} \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y}_r &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T (\boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}} \boldsymbol{A}_{\mathcal{T}^{l-1}}^{\dagger} \boldsymbol{y}) \\ &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{A}_{\mathcal{T}^{l-1}} \left((\boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{A}_{\mathcal{T}^{l-1}})^{-1} \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \right) \boldsymbol{y} \\ &= \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^{l-1}}^T \boldsymbol{y} = \boldsymbol{0}. \end{aligned}$$

Therefore, $i_l = \arg\max_i |\langle \boldsymbol{a}_i, \boldsymbol{y}_r \rangle| \notin \mathcal{T}^{l-1}$. [4]

Comments

Some students got the basic idea but failed to give a mathematically sounded proof.

iii A. The mutual coherence of $\boldsymbol{A}$ is defined as

$$\mu(\boldsymbol{A}) = \max_{i \neq j} |\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle|.$$

[1]

B.

$$|\langle \boldsymbol{a}_i, \boldsymbol{y} \rangle| = \left| \boldsymbol{a}_i^T \left(\sum_{j \in \mathcal{T}_0} \boldsymbol{a}_j \boldsymbol{x}_{0,j} \right) \right| \overset{(a)}{\leq} \sum_{j \in \mathcal{T}_0} \left| \boldsymbol{a}_i^T \boldsymbol{a}_j \right| |\boldsymbol{x}_{0,j}|$$
$$\overset{(b)}{\leq} \mu \sum_{j \in \mathcal{T}_0} |\boldsymbol{x}_{0,j}| \overset{(c)}{\leq} \mu \sqrt{S} \left\| \boldsymbol{x}_0 \right\|_2,$$

where (a) comes from triangle inequality, (b) follows from the definition of mutual coherence, and (c) holds due to Cauchy-Schwartz inequality. [3]

(b) i The complete SP algorithm is stated in Algorithm 2.4.

Algorithm 2.4 SP Algorithm

Input: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
Initialisation: $\mathcal{T}^0 = \text{supp}(H_S(\boldsymbol{A}^T \boldsymbol{y}))$, and $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^0} \boldsymbol{A}_{\mathcal{T}^0}^{\dagger} \boldsymbol{y}$.
while $l = 1, 2, \cdots$ until stop criteria hold **do**
  $\tilde{\mathcal{T}}^l = \mathcal{T}^{l-1} \bigcup \text{supp}(H_S(\boldsymbol{A}^T \boldsymbol{y}_r))$ ▷ Expand support
  Set $\boldsymbol{b}$ by $\boldsymbol{b}_{\tilde{\mathcal{T}}^l} = \boldsymbol{A}_{\tilde{\mathcal{T}}^l}^{\dagger} \boldsymbol{y}$ and $\boldsymbol{b}_{\left(\tilde{\mathcal{T}}^l\right)^c} = \boldsymbol{0}$ ▷ Estimate a $2S$-sparse signal
  Set $\mathcal{T}^l = \text{supp}(H_S(\boldsymbol{b}))$ ▷ Shrink support
  Set $\boldsymbol{x}$ by $\boldsymbol{x}_{\mathcal{T}^l} = \boldsymbol{A}_{\mathcal{T}^l}^{\dagger} \boldsymbol{y}$ and $\boldsymbol{x}_{\left(\mathcal{T}^l\right)^c} = \boldsymbol{0}$ ▷ Estimate an S-sparse signal
  $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

[2]

ii
$$H_3([-9, 8, -7, 6, -5]^T) = [-9, 8, -7, 0, 0].$$

[2]

iii A matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is said to satisfy the RIP with parameters (K, δ), if for all $\mathcal{T} \subset [n]$ such that $|\mathcal{T}| \leq K$ and for all $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|}$, it holds that

$$(1 - \delta) \left\| \boldsymbol{q} \right\|_2^2 \leq \left\| \boldsymbol{A}_{\mathcal{T}} \boldsymbol{q} \right\|_2^2 \leq (1 + \delta) \left\| \boldsymbol{q} \right\|_2^2.$$

The RIC δ_K is defined as the smallest constant δ for which the K-RIP holds, i.e.,

$$\delta_K = \inf \left\{ \delta : \ (1 - \delta) \left\| \boldsymbol{q} \right\|_2^2 \leq \left\| \boldsymbol{A}_{\mathcal{T}} \boldsymbol{q} \right\|_2^2 \leq (1 + \delta) \left\| \boldsymbol{q} \right\|_2^2. \right.$$
$$\left. \forall\, |\mathcal{T}| \leq K, \ \forall \boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|} \right\}.$$

[4]

iv $-\delta_{2S} \leq \sigma\left(\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{J}}\right) \leq \delta_{2S}$ where $\sigma\left(\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{J}}\right)$ denotes the singular value of the matrix $\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{J}}$. [2]

Solution to Problem 3.

(a) i As the constraints in (3.1) involves m many equations, it is clear that $\boldsymbol{v} \in \mathbb{R}^m$. The Lagrangian is given by

$$L(\boldsymbol{x}, \boldsymbol{v}) = \frac{1}{2} \|\boldsymbol{x}\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}).$$

[2]

ii The dual function is given by

$$g(\boldsymbol{v}) = \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v}) = \min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{x}\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}).$$

To find the closed form of $g(\boldsymbol{v})$, set $\frac{d}{d\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v})$ to $\mathbf{0}$. Note that

$$\frac{d}{d\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{x} - \boldsymbol{A}^T\boldsymbol{v}.$$

It is clear that the optimal

$$\boldsymbol{x}^\star = \boldsymbol{A}^T\boldsymbol{v}.$$

Hence

$$\begin{aligned} g(\boldsymbol{v}) &= \frac{1}{2} \left\|\boldsymbol{A}^T\boldsymbol{v}\right\|_2^2 + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}) \\ &= \boldsymbol{v}^T\boldsymbol{y} - \frac{1}{2}\boldsymbol{v}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}. \end{aligned}$$

[2]

iii The dual problem is given by

$$\max_{\boldsymbol{v}} g(\boldsymbol{v}) = \max_{\boldsymbol{v}} \boldsymbol{v}^T\boldsymbol{y} - \frac{1}{2}\boldsymbol{v}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}.$$

By setting $\frac{d}{d\boldsymbol{v}}\ g(\boldsymbol{v})$ to $\mathbf{0}$, one obtains

$$-\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v}^\star + \boldsymbol{y} = \mathbf{0}.$$

Or equivalently

$$\boldsymbol{v}^\star = (\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{y}.$$

[2]

iv From the above steps, it is clear that

$$\boldsymbol{x}^\star = \boldsymbol{A}^T\boldsymbol{v}^\star = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1}\boldsymbol{y}\ (= \boldsymbol{A}^\dagger\boldsymbol{y}).$$

[2]

Comments

Some students made mistakes in linear algebra. For example, mistakes in computing $\frac{d}{d\boldsymbol{x}}L(\boldsymbol{x},\boldsymbol{v})$ and $\frac{d}{d\boldsymbol{v}}g(\boldsymbol{v})$, not realising $\boldsymbol{v}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{v} = \|\boldsymbol{A}^T\boldsymbol{v}\|_2^2$, and the mistaken expression $\frac{y}{\boldsymbol{A}^2}$.

(b) i It is clear that

$$\partial_x f(x) = x - y + \lambda \cdot \operatorname{sign}(x),$$

where

$$\operatorname{sign}(x) = \begin{cases} 1 & \text{if } x > 0, \\ [-1,1] & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Hence, it is straightforward to verify that

$$x^\star = \eta_\lambda(y) = \begin{cases} y - \lambda & \text{if } y > \lambda, \\ 0 & \text{if } y \in [-\lambda, \lambda], \\ y + \lambda & \text{if } y < -\lambda. \end{cases}$$

[4]

ii It is clear that

$$\nabla_{\boldsymbol{x}} g(\boldsymbol{x})|_{\boldsymbol{x}^k} = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y}).$$

It holds that

$$\begin{aligned} &\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \\ &\approx \frac{1}{2}\left\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^k\right\|_2^2 + \left\langle \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y}), \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \\ &= c + \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k + t_k\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1. \end{aligned}$$

The optimal solution of

$$\min_{\boldsymbol{x}} \frac{1}{2t_k}\left\|\boldsymbol{x} - \boldsymbol{x}^k + t_k\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})\right\|_2^2 + \lambda\|\boldsymbol{x}\|_1$$

admits the closed form

$$\boldsymbol{x}^\star = \eta_{\lambda t_k}(\boldsymbol{x}^k - t_k \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})).$$

Hence, the IST algorithm is given by the following iteration:

$$\boldsymbol{x}^{k+1} = \eta_{\lambda t_k}(\boldsymbol{x}^k - t_k \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{x}^k - \boldsymbol{y})).$$

[3]

(c) i It is clear that (3.6) can be written as

$$\min_{\boldsymbol{x},\boldsymbol{z}} \|\boldsymbol{z}\|_1 \text{ s.t. } \begin{bmatrix} \boldsymbol{D} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} \boldsymbol{0} \\ -\boldsymbol{I} \end{bmatrix} \boldsymbol{z} = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{0} \end{bmatrix}.$$

Hence, $f(\boldsymbol{x}) = 0$, $g(\boldsymbol{z}) = \|\boldsymbol{z}\|_1$,

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{D} \\ \boldsymbol{I} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0} \\ -\boldsymbol{I} \end{bmatrix}, \text{ and } \boldsymbol{c} = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{0} \end{bmatrix}.$$

[3]

ii

$$\begin{aligned} L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v}) = &\|\boldsymbol{z}\|_1 + \boldsymbol{v}^T(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}) \\ &+ \frac{\rho}{2}\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_2^2, \end{aligned}$$

where $\boldsymbol{v} \in \mathbb{C}^{m+n}$. [3]

iii It is straightforward to verify that

$$L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v}) = c + \|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} + \frac{1}{\rho}\boldsymbol{v}\right\|_2^2.$$

Then,

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} L_\rho(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{v}^k) \\ &= \boldsymbol{A}^\dagger\left(-\boldsymbol{B}\boldsymbol{z}^k + \boldsymbol{c} - \frac{1}{\rho}\boldsymbol{v}^k\right), \end{aligned}$$

and

$$
\begin{aligned}
\boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}} L_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{v}^k) \\
&= \arg\min_{\boldsymbol{z}} \ \|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|-\boldsymbol{z} + \frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right\|_2^2 \\
&= \arg\min_{\boldsymbol{z}} \ \frac{1}{\rho}\|\boldsymbol{z}\|_1 + \frac{1}{2}\left\|\boldsymbol{z} - \frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right\|_2^2 \\
&= \eta_{\frac{1}{\rho}}\left(\frac{1}{\rho}\boldsymbol{v}^k_{m+1:m+n}\right).
\end{aligned} \tag{3.9}
$$

[4]

Comments

Equation (3.9) is the key step to find the closed form solution for updating $\boldsymbol{z}$. It uses the special structure of the matrix $\boldsymbol{B}$.